



Illuminating patterns of firefly abundance using citizen science data and machine learning models

Darin J. McNeil^{a,*}, Sarah C. Goslee^b, Melanie Kammerer^b, Sarah E. Lower^c, John F. Tooker^d, Christina M. Grozinger^d

^a Department of Forestry and Natural Resources, University of Kentucky, Lexington, KY 40506, USA

^b United States Department of Agriculture – Agricultural Research Service, Pasture Systems and Watershed Management Research Unit, University Park, PA 16802, USA

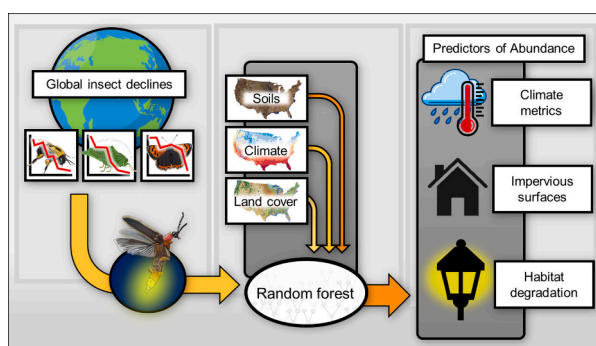
^c Department of Biology, Bucknell University, Lewisburg, PA 17837, USA

^d Department of Entomology, Insect Biodiversity Center, Huck Institutes of the Life Sciences, Pennsylvania State University, University Park, PA 16802, USA

HIGHLIGHTS

- Identifying factors driving declines in insects is critical for their conservation.
- We used citizen science data on fireflies to study a suite of ecological patterns.
- Firefly abundance was driven by soil conditions, climate/weather, and land cover.
- Stressors appeared to be context-specific with respect to habitat and climate.
- Climactic patterns indicated that firefly populations will be influenced by climate change.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Rafael Mateo

Keywords:

Climate
Ecology
Firefly
Insect conservation
Lampyridae
Lightning bug

ABSTRACT

As insect populations decline in many regions, conservation biologists are increasingly tasked with identifying factors that threaten insect species and developing effective strategies for their conservation. One insect group of global conservation concern are fireflies (Coleoptera: Lampyridae). Although quantitative data on firefly populations are lacking for most species, anecdotal reports suggest that some firefly populations have declined in recent decades. Researchers have hypothesized that North American firefly populations are most threatened by habitat loss, pesticide use, and light pollution, but the importance of these factors in shaping firefly populations has not been rigorously examined at broad spatial scales. Using data from >24,000 surveys (spanning 2008–16) from the citizen science program Firefly Watch, we trained machine learning models to evaluate the relative importance of a variety of factors on bioluminescent firefly populations: pesticides, artificial lights at night, land cover, soil/topography, short-term weather, and long-term climate. Our analyses revealed that firefly abundance was driven by complex interactions among soil conditions (e.g., percent sand composition), climate/weather (e.g., growing degree days), and land cover characteristics (e.g., percent agriculture and impervious cover). Given the significant impact that climatic and weather conditions have on firefly abundance, there is a strong likelihood that firefly populations will be influenced by climate change, with some regions becoming higher quality

* Corresponding author.

E-mail address: darin.j.mcneil@uky.edu (D.J. McNeil).

<https://doi.org/10.1016/j.scitotenv.2024.172329>

Received 28 September 2023; Received in revised form 6 April 2024; Accepted 6 April 2024

Available online 10 April 2024

0048-9697/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

and supporting larger firefly populations, and others potentially losing populations altogether. Collectively, our results support hypotheses related to factors threatening firefly populations, especially habitat loss, and suggest that climate change may pose a greater threat than appreciated in previous assessments. Thus, future conservation of North American firefly populations will depend upon 1) consistent and continued monitoring of populations via programs like Firefly Watch, 2) efforts to mitigate the impacts of climate change, and 3) insect-friendly conservation practices.

1. Introduction

In recent decades, scientists have increasingly reported precipitous, widespread declines in many global insect populations (Hallmann et al., 2017; Van Klink et al., 2020; Wagner et al., 2021a). These declines do not seem to be restricted to a particular region (Høye et al., 2013; Sands, 2018; Janzen and Hallwachs, 2019; Wepprich et al., 2019), insect taxon (Wagner, 2020), or even insect functional group (Harvey et al., 2020). Indeed, among the many documented population losses are: butterflies (Thomas, 2016; Forister et al., 2021; Van Swaay et al., 2022), bees (Potts et al., 2010), grasshoppers (Dirzo et al., 2014), beetles (Brooks et al., 2012; Homburg et al., 2019; Hallmann et al., 2020), and others (Høye et al., 2013; Stepanian et al., 2020). Declines of global insect populations are troubling because insects may broadly serve as indicators of environmental degradation (Brown, 1997) and are foundational to ecosystem function, including trophic interactions (Wagner, 2020), and ecosystem services (Kevan and Viana, 2003). While there have been voices of skepticism regarding insect declines (Crossley et al., 2020; Neff et al., 2022), the issue has been deemed critical enough that local, regional, and federal policies and incentive programs have been developed to help stem apparent insect declines (Forister et al., 2019), though conservation biologists emphasize that there remain enormous unknowns that must be addressed before we can effectively conserve many populations (Harvey et al., 2020).

For most insect groups that have been evaluated, the factors contributing to population declines are varied (Van Klink et al., 2020; Raven and Wagner, 2021). Perhaps the single most cited driver of population losses in insects is land conversion, especially conversion to agriculture and urban cover (Wilcove et al., 1998; Raven and Wagner, 2021). Land conversion is one of the most conceptually straightforward drivers of insect declines because it removes habitat necessary to sustain insect populations (Wagner, 2020). In the case of agricultural intensification, landscapes themselves can be severely degraded by loss of available habitat (Brooks et al., 2002; Outhwaite et al., 2022), but some insects can also be harmed by local crop management, like tillage or application of pesticides (mostly insecticides and herbicides) that cause indirect and direct mortality (Forister et al., 2019; Olaya-Arenas et al., 2020). In fact, a recent assessment of insecticide use across the United States indicated that, although the absolute quantity of insecticide applied in agricultural landscapes has declined over the past 20 years, toxic loads have increased ninefold due to the increasing prevalence of neonicotinoid insecticides, which can be toxic to insects at very low doses (Douglas et al., 2020). Weather and climate conditions, which are subject to effects of climate change, are also known to be major drivers of variation in insect abundance and diversity (Filazzola et al., 2021; Halsch et al., 2021; Harvey et al., 2023; Müller et al., 2023). Weather conditions, in particular, have complex direct and indirect effects on insects, by altering patterns of phenology, exposing them to environmental extremes to which they are not physiologically adapted, or causing phenological mismatch (Raven and Wagner, 2021). Lastly, invasive species can change habitat conditions or cause disease, which can drive populations and species declines as well (Wagner and Van Driesche, 2010). Importantly, these stressors can act synergistically, or have unpredictable outcomes (Forister et al., 2019).

One insect group of global conservation concern are the fireflies (Coleoptera: Lampyridae; Lewis et al., 2020; Mbugua et al., 2020). With over 2000 described species across the globe, this beetle family includes

a variety of species, some that produce light as adults (primarily to attract mates) and others that do not, though some of these species produce light as eggs, and all produce light as larvae (Lewis, 2016; Faust, 2017). The species that bioluminesce as adults (hereafter, “fireflies”) are regarded as highly charismatic (Lewis et al., 2021) and their predaceous, soil-dwelling larvae may have the capacity to perform the key ecological service of consuming agricultural pests (McLean et al., 1972; Barrows et al., 2008). Although quantitative data on firefly populations are not available for most species (Fallon et al., 2021), a growing number of anecdotal reports suggest that, in some regions, firefly populations have declined in recent decades (Picchi et al., 2013; Chatragadda, 2020; Mbugua et al., 2020). However, as with most groups of insects, there is likely to be spatial heterogeneity in population trends (e.g., Asia vs. North America [Chatragadda, 2020; Evans et al., 2019], also see Wagner et al., 2021b). A recent assessment of potential drivers of declines around the world proposed possible threats to fireflies (Lewis et al., 2020), but empirical assessments do not exist for the overwhelming majority of individual species, and the roles of these potential drivers remain largely hypothetical. To make matters more complicated, different species all have different phenologies, mating times, and habitat preferences (Fallon et al., 2019; Reed et al., 2020), making large-scale studies challenging. Recent research examined the role of climate on North American firefly populations and found that winter and spring maximum temperatures and soil moisture were significant drivers of abundance (Evans et al., 2019). Lewis et al. (2020) suggested additional weather factors and land use conditions with strong potential to influence firefly populations, as is true for many insect populations that appear to be in decline (Van Klink et al., 2020; Wagner et al., 2021a). However, as in many other wide-ranging insect groups, quantitative investigation of these hypotheses in fireflies has been hampered by a dearth of large, geographic-scale, population data (Fallon et al., 2021).

One potential source of biological information to inform firefly conservation is citizen science projects (Dickinson et al., 2010). Indeed, well-designed citizen science programs can accrue large quantities of data across broad geographic extents, yielding volumes of data that would otherwise be nearly impossible to collect with traditional field-based sampling methods (Tulloch et al., 2013; McKinley et al., 2017; Bradter et al., 2018). For instance, Cornell Lab of Ornithology’s eBird enterprise has become the world’s largest database of avian presence information (Sullivan et al., 2014) and they have now produced lifecycle-wide population range maps, habitat associations, and even population trend maps for many bird species (Fink et al., 2020). Given that citizen science data are well suited for charismatic species like birds and fireflies (Dickinson et al., 2010; Lewis et al., 2021), it is no surprise that a variety of citizen science programs have been developed to help gather occurrence data for adult lampyrids and increase awareness about their bionomics and conservation status: ‘Light Up West Virginia’ (wvdnr.gov/fireflies), ‘Glow Up!’ (Picchi et al., 2013), ‘UK Glowworm Survey’ (glowworms.org.uk), Firefly Atlas (Xerces Society, fireflyatlas.org), and ‘Firefly Watch’ (massaudubon.org), among numerous others.

Despite the growing number of citizen science programs aimed at fireflies, few broad-scale analyses have been conducted to describe the ecological needs of these charismatic beetles. A recent assessment surveyed firefly experts from across the globe to rank the relative threats to worldwide firefly populations (Lewis et al., 2020); their analysis concluded that in North America the greatest threats to firefly populations are: 1) habitat loss, 2) pesticide exposure, 3) light pollution, and

4) climate-driven factors (e.g., drought, storms), in that order. Although the assessment published by Lewis et al. (2020) provided useful hypotheses on the threats to firefly populations, formal ecological analyses quantifying the relative impacts of these and other stressors on fireflies remain critical for effective conservation (Cardoso et al., 2011). To quantify the varied ecological needs of wild populations of fireflies, we evaluated how patterns of firefly abundance varied as a function of both these hypothesized factors and a suite of landscape/climate features across eastern North America. More specifically, we correlated abundance data for luminescent species from the citizen science program ‘Firefly Watch’ with a variety of remote-sensed data including land cover, light pollution, pesticide expected toxic load, soil characteristics, localized weather, and regional climate. To accomplish this, we leveraged the machine learning method ‘random forest’ to assess the relative value of various co-linear landscape indices in their capacity to explain variation in local firefly abundance. We further discuss how our findings apply to North American firefly conservation and responses to landscape stressors.

2. Materials and methods

2.1. Firefly Watch dataset

The Firefly Watch program was initiated in 2008 through the Museum of Science in Boston, Massachusetts (U.S.A.) to describe the phenology, distribution, population trends, and ecology of fireflies. In 2017, Firefly Watch was moved from the Museum of Science to Massachusetts Audubon, who currently maintain the archived data and promote the program. The Massachusetts Audubon website includes the raw Firefly Watch observations dataset as well as instructions for observers on how to most appropriately conduct firefly surveys. In particular, observers are instructed to conduct ten-second surveys wherein observers record the number of flashing individual fireflies, which may represent multiple species signaling concurrently in an area. Observers make coarse attempts to quantify the number of firefly species, though not the counts per “species”. In addition to firefly observations, surveyors are asked to record a variety of information about the survey conditions (e.g., temperature, cloud cover, etc.).

To process the Firefly Watch dataset for analyses, we downloaded the dataset from massaudubon.org and filtered it to use data from 2008 (the first year) through 2016; although Firefly Watch data exist for 2017 until the present, firefly count categories were changed after 2016, complicating analyses of all data together. All analyses presented here were conducted in Program R (R Core Team, 2023). In processing the Firefly Watch data, we used the reported coordinate locations to extract survey time zones using the *lutz* package in program R (Teucher and Rudis, 2019). Incorporating time zones and user-reported survey times, we calculated site-specific sunset times and, thereby, minutes-since-sunset using the *lubridate* package (Spinu et al., 2021). Minutes-since-sunset provided a standardized measure of time within a single day that was more straightforward to model than raw survey time. We also converted raw, user-reported dates to ordinal date for analyses because ordinal date can be modeled as a linear variable. Finally, as a response variable, we used the user-reported firefly count. However, Firefly Watch users report abundance in one of five bins: 0, 1, 2–5, 6–20, and > 20. To linearize these categories, we used the lowest possible count in each bin as a ‘minimum count’ for each survey. Therefore, counts for the respective categories were: 0, 1, 2, 6, and 21. Although this treatment of count categories is analytically appropriate, absolute magnitudes of effect sizes should be interpreted as conservative because we used a minimum possible count. Because our focus was on the eastern United States, we only used Firefly Watch data from the conterminous US, east of the 100th meridian (Fig. 1).

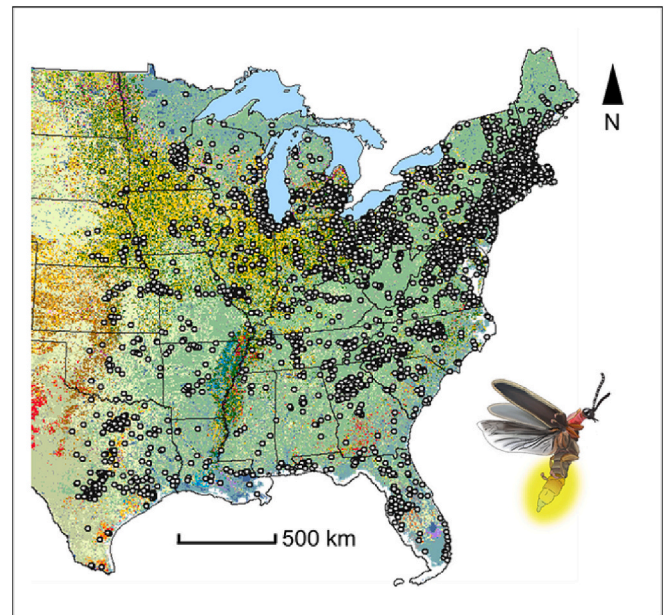


Fig. 1. A map depicting Firefly Watch survey locations (white circles) from 2008 to 2016 across the eastern United States overlaid upon the US Cropland Data Layer 2016.

2.2. Land-use variables

To quantify habitat features, we extracted the most common land cover metrics represented at firefly sampling locations. First, we used the USDA Cropland Data Layer (USDA National Agricultural Statistics Service, 2016) to calculate percent cover for the following variables: barren, grass/pasture, open water, row-crop, and shrubland. We also combined several variables into relevant composite variables: emergent wetland + woody wetland = ‘percent wetland’, coniferous + deciduous + mixed forest = ‘percent forest’, and crops + pasture = ‘percent agriculture’. To quantify human development, we used the Urban Imperviousness layers available from the National Land Cover Database (Yang et al., 2003; Xian et al., 2011). Finally, to approximate effects of light pollution on firefly counts, we extracted digital number (DN) values from the Harmonized Global Nighttime Lights dataset (1992–2018) presented by Li et al. (2020).

To estimate insecticide toxic load around each firefly sampling location, we used the insecticide reclass tables presented by Douglas et al. (2020). This insecticide index was generated using year-, state-, and crop-specific loads of all insecticides applied to each cover type present within the Cropland Data Layer, each scaled by the toxicity to a model insect species, the honey bee (*Apis mellifera*). We rescaled the Cropland Data Layer using the reclass table presented by Douglas et al. (2020) via the SpeedyBeeModel published by Kammerer et al. (2021). Each location was weighted by 500 m around each firefly location. A more detailed description of the pesticide extraction process can be found within Douglas et al. (2020, 2022) and McNeil et al. (2020).

2.3. Climate and weather variables

To quantify weather and climate variables across the eastern United States, we generated a set of standard bioclimatic variables using the daily 4-km gridded weather data from PRISM (PRISM Climate Group, 2014). The first data type we used was a subset of those from the standard BIOCLIM dataset commonly used to predict occurrence patterns for plants and animals (Fick and Hijmans, 2017). We selected standard BIOCLIM variables, subset from the complete published set of BIOCLIM variables based on their relative value in other relevant modeling exercises (e.g., Wang et al., 2017; Calovi et al., 2021) while excluding the

most highly correlated variables and making sure each type of climate variable is represented. For each variable, we extracted quantiles (either the 10th or 90th percentile, as most appropriate) rather than the mean, because extremes are often more biologically relevant than statistical normals. The BIOCLIM variables included in our analyses were isothermality (bc03), temperature seasonality (bc04), various measures of temperature (e.g., maximum, minimum, range, etc.: bc02, bc05, bc06, bc07, bc08, and bc09) and various measures of precipitation (e.g., precipitation in wettest month, precipitation seasonality, etc.: bc13, bc15, bc17, bc18, and bc19; see Table A.1). Additionally, we extracted several relevant variables detailed by the USDA – Natural Resources Conservation Service (NRCS) Range and Pasture Handbook (NRPH; Natural Resources Conservation Service, 2003). These variables included number of frost-free days (-2.2°C), growing degree days (base 4.4°C), annual precipitation, and maximum number of consecutive days without precipitation of at least 0.25 cm. Growing degree days (hereafter, GDDs), is defined as the accumulation of degrees above a base temperature (4.4°C in this study) over a growing season. Variables were extracted for the following temporal periods: quantiles for 2001–2015 (mean), variables for the year in which each firefly survey was conducted, variables for the year prior to which each firefly survey was conducted (previous year), and variables for two years prior to each firefly survey (two years prior; Evans et al., 2019). This was done because many eastern North American firefly species are thought to have a 1–2 year generation time (Faust, 2017) and the annual emergence of many firefly species can be predicted with GDDs (Faust, 2012). We also calculated the GDDs at the time of survey for each point.

2.4. Soil and topographic variables

Climate data were complemented with finer-resolution soils and topographic variables. Soils data included a suite of relevant variables previously established to predict biological responses by plants and insects (e.g., Wang et al., 2017). In particular, we included the following soil variables in our firefly analyses: root zone depth, National Commodity Crop Productivity Index (all crops), soil organic carbon, soil available water storage, percent earthy components, and soil bulk density. We also included soil composition data: percent sand and percent clay (percent silt was not included because it was redundant with sand and clay). These data were obtained from gSSURGO for 0–100 cm (USDA Natural Resources Conservation Service, Soil Survey Staff, 2020). Finally, we included two important topographic variables potentially relevant to firefly abundance due to influence on micro-habitat: aspect (east-west) and slope (degrees). These were calculated from the 1/3 arc-second (approximately 30 m) 3DEP digital elevation model (US Geologic Survey, 2019).

2.5. Random forest analyses

To model firefly responses to a suite of ecological factors (survey conditions, land use, soils, weather, and climate), we used the machine learning method random forest. Random forest is ideal for exploring patterns in large datasets such as those considered here with the Firefly Watch dataset because random forest models can make robust, non-linear inferences from datasets that may include correlated predictors (Wright et al., 2020). For each random forest model in our analysis, we used minimum firefly count (see “Firefly Watch dataset” above) as a function of relevant predictor variables. All random forest models were implemented using the *ranger* package in R (Wright and Ziegler, 2017). Preliminary parameter tuning suggested that root mean squared error was constant across a wide span of *mtry* (maximum number of variables split at each node) and tree numbers so we used the default *mtry* (which varied for each analysis) and set *ntrees* (number of trees in each ‘forest’) to a constant level = 5000. After fitting each random forest model, we reviewed functional relationships using partial dependence plots, generated using the *partial* function within the *pdp* package (Greenwell,

2018). We visualized each response between the 5th and 95th percentile of each predictor variable to avoid predicting into data-poor portions of our model. We use the mean response across each of these predictions as a benchmark reference. For each model, we also calculated root mean squared error as a measure of model performance, except for our detection model (discussed below) which was binary; for detection, we calculated Brier’s Score as a model performance metric (Wright and Ziegler, 2017).

To assess how different variables influenced the drivers of heterogeneity in firefly counts (including the observation process), we developed a hierarchical set of random forest models. Specifically, we developed models in three model Tiers.

Tier 1 models. The first model tier (Tier 1) included two individual models to predict firefly counts: one to assess the role of survey covariates (e.g., ‘minutes since sunset’ at survey time, Model Tier 1a, hereafter, ‘detection model’) and another (Tier 1b, hereafter, ‘climate model’) to assess broad climate suitability. The detection model consisted of a binary outcome (fireflies counted [1] or no fireflies counted [0]) as a function of seven survey covariates recorded by firefly watch participants at the time of survey: ordinal date, minutes-since-sunset, temperature at the time of survey, precipitation index, windspeed index, cloudiness index, and moonlight index. We used a binary outcome for the detection model because we were primarily interested in modeling whether survey conditions were conducive for firefly activity or not. The climate model, in contrast, used aggregated minimum firefly counts using the same 4 km grid as the PRISM climate data; mean minimum count/grid cell was modeled as a function of grid cell climate conditions.

Using the detection model, we predicted the probability of detecting fireflies during each survey (using the *predict.ranger* function), providing us a model-based estimate for survey detection conditions (values close to 0 indicating poor sampling conditions and values close to 1 excellent sampling conditions). This ‘detection score’ was used in all Tier 2–3 models as a weight that down-weighted samples from poor survey conditions and allocated greater weight to surveys conducted during conditions good for firefly display (e.g., within the best date range, time window, temperature conditions, etc.). We also predicted a ‘climate score’ at each location using a similar procedure and applied this climate score to each model as a covariate to account for the variation in firefly response driven by climate patterns. The climate model included the 15-year quantiles for the following variables: number of frost-free days, number of GDDs, total precipitation, maximum number of consecutive days without precipitation, and BIOCLIM variables b02–b09, b13, b15, and b17–b19.

Tier 2 model. After modeling detection and broad climate patterns, we created a single Tier 2 model that incorporated all other covariates while accounting for detection probability (using weights) and broad climatic suitability (as a climate score, treated as a model variable). This Tier 2 model included 79 variables: the climate score, 10 soil variables, 11 land cover (night-time lights, pesticide load, and nine variables from CDL), and 58 weather variables.

The model incorporated the following soil variables: 1. available water storage, 2. percent clay, 3. bulk density, 4. percent sand, 5. soil organic carbon, 6. combined crop index, 7. percent earth, 8. root zone depth, 9. topographic slope, and 10. topographic aspect.

Land cover variables were as follows: 1. Night Time Lights “digital number” (DN), 2. Pesticide toxic load, and percent cover for CDL variables within 500 m of each survey location: 3. Open water cover, 4. Barren cover, 5. Shrubland cover, 6. Grass/pasture cover, 7. Row crop cover, 8. Forest cover, 9. Wetland cover, 10. Agriculture cover, and 11. Impervious surface cover.

We included coalesced 58 weather variables known to be (or hypothesized to be) relevant to fireflies in some contexts (Evans et al., 2019; Lewis et al., 2020): GDDs (at time of survey), spring precipitation, spring minimum temperature, spring maximum temperature, winter precipitation, winter minimum temperature, and winter maximum

temperature. Based on published literature on relationships between firefly life history and weather, we included data representing the survey year, year prior, and two years prior for many variables: number of frost-free days, number of GDDs, total precipitation, maximum number of consecutive days without precipitation, and the following BIOCLIM variables: b02-b09, b13, b15, and b17-b19.

Tier 3 models. We specified four Tier 3 models, each of which was constructed using the same variables and general procedure as the Tier 2 model but trained using different subsets of the data. Two of the Tier 3 models were intended to examine variables important to firefly count in ‘high-quality’ climates (i.e., high climate score) and ‘low-quality’ climates (i.e., low climate score). We defined ‘high-quality climates’ as the highest 1/3 quantile of the climate score and ‘low-quality climates’ as the lowest 1/3 climate quantile. The final two models in Tier 3 were analogous to the high/low-quality climate models but conducted using impervious cover – highest 1/3 quantile treated as ‘low-quality habitats’ (high impervious cover = low-quality) and lower 1/3 quantile interpreted as ‘high-quality habitats’ (low impervious cover = high-quality). We mapped our Tier 3 model over an example landscape that contains a diversity of land cover types, Centre County, Pennsylvania with year 2016 land cover/climate data and mean values for weather variables.

3. Results

The 2008–16 Firefly Watch dataset for the United States east of the 100th meridian yielded 24,683 usable surveys for assessing patterns of firefly occurrence (Fig. 1, Table 1). The number of surveys varied among years with the lowest occurring in the first year (2008; $n = 1811$) and the maximum occurring in 2011 ($n = 3403$). The mean minimum count of fireflies was 4.51 individuals/survey across all years and was lowest in the first year (3.41 individuals/survey) and highest in 2015 (4.97 individuals/survey). Of the 24,683 surveys in the data subset we analyzed, 20,194 (82 %) included a count that was non-zero (at least one firefly reported). Surveys were conducted in a wide variety of ecological contexts; most surveys had at least some forest cover within 500 m ($\bar{x} = 23.50$ %, $SD = 23.09$) with 6778 surveys (27 %) occurring with <5 % forest cover and 3714 surveys (15 %) occurring with >50 % forest cover. Similarly, surveys were varied in the extent of urbanization (i.e., percent impervious cover) with most surveys including at least some impervious cover ($\bar{x} = 18.21$ %, $SD = 16.87$). One third of surveys ($n = 8081$; 33 %) had <5 % impervious cover and only 5 % ($n = 1321$) occurred within habitats dominated by >50 % impervious cover (Table A.1).

In our Tier 1 models, the top detection variables were ordinal date and minutes since sunset (Brier Score = 0.12; Table 1, Table A.2); on

average, the probability of observing a firefly was maximized in late July and for counts that occurred between 30 min pre-sunset and about 60 min post-sunset (Fig. 2A-B). Our Tier 1 climate model indicated that the most important predictors of counts was average season-wide GDDs and minimum temperature of the coldest month (bc06; $R^2 = 0.03$; Table 1, Table A.2). Over the course of a year, firefly abundance was generally, negatively affected by increased GDDs with the exception of a pronounced dip around 2700 GDDs (Fig. 2C-D). When this climate model was mapped across our study area, regions predicted to have the greatest firefly counts were 1) the Midwest/Great Lakes region, 2) the upper Mid-Atlantic region, 3) the Red Hills region of the Southeast, and 4) portions of the Great Plains (Fig. 2E).

After accounting for the effects of detection and broad climate, our combined model, which used all 24,683 samples (Tier 2, $R^2 = 0.39$; Table 1), indicated that a variety of factors influenced firefly abundance with weather conditions (e.g., GDDs [accumulated at the time of survey and average accumulated GDD among years], soil conditions (percent sand), and land-cover characteristics (% impervious cover, % agriculture cover, and night-time lights) all within the top ten variables (Table 2). Regarding short-term weather patterns, functional relationships indicated that there was a positive relationship between the number of GDD in the survey year and firefly counts (Fig. 3); however, there was a threshold within a year, beyond which counts declined (i.e., beyond ~1400 accumulated GDD YTD; Fig. 3). The GDD covariate was also strongly supported for both previous year and two years prior, however, the effect size was small (Table 2, Fig. 3). Firefly counts were also negatively related to percent soil composition that was sand (Fig. 3). Similarly, firefly abundance showed a negative relationship with percent impervious cover with a pronounced decline occurring beyond 5–10 % cover (Fig. 3). Firefly abundance was also negatively related to nighttime lights DN (Table 2, Fig. 3). Finally, firefly counts were positively associated with the percent area of agriculture cover within 500 m (Fig. 3). A full listing of the 79 variables used in our models and their relative importance to predicting firefly counts is presented in Table A.3.

Based on past literature (e.g., Lewis et al., 2020), we expected several variables to be strongly predictive of firefly abundance but they were not among the top variable importance scores. In particular, metrics of precipitation did not appear in the top models (e.g., winter precipitation was the 34th ranked variable and all other precipitation variables were lower) nor did metrics of pesticide load (Table 2). The functional relationship between firefly abundance and pesticide load, though largely flat, suggested a slight increase from 0 to lower levels of pesticide toxic load but this was almost certainly due to the inherent correlation between pesticide toxic load and percent agriculture cover. When we mapped our model across an example landscape, Centre County Pennsylvania, firefly density was estimated to be highest within agricultural valleys, lowest in highly urban areas, and intermediate within suburbs, forest, and ecotones (Fig. 4).

We next conducted Tier 3 models evaluating firefly counts in low/poor and high/good climatic conditions. Our “low climate score” subset model ($n = 8526$ surveys) had $R^2 = 0.21$ (Table 1). The top variables within this “low climate score” subset model were i. accumulated GDD at the time of survey, ii. percent sand, and iii. Previous year’s total GDDs (Table 2). In contrast, our “high climate score” subset model ($n = 8152$ surveys) had $R^2 = 0.27$. Top variables within this “high climate score” subset model were i. accumulated GDD at the time of survey, ii. percent impervious surfaces, and iii. Percent forest (Table 2, see Table A.3 for results from all variables).

In our Tier 3 models evaluating the impact of habitat quality, when we subset the data by percent impervious cover, our “high impervious cover” (low-quality habitat) subset model ($n = 8148$ surveys) had $R^2 = 0.33$ (Table 1). The top variables within the “high impervious cover” subset model were i. accumulated GDD at the time of survey, ii. percent clay, and iii. Percent sand. Similarly, our “low impervious cover” (high-quality habitat) subset model ($n = 8150$ surveys) had $R^2 = 0.38$. Top variables within this “low impervious cover” subset model were i.

Table 1

Model performance metrics for all 10 random forest models of the Firefly Watch dataset. For each model, we report response variable, dataset, error (Out-of-bag mean squared error except the detection model which is Brier score), R^2 , number of predictors sampled at split (mtry), number of model parameters (k), and sample size (N). All models were run with 5000 trees.

Model	Response variable	Dataset	Error	R^2	mtry	k	N
1a	Presence (1/0)	Complete	0.12	–	2	7	24,683
1b	Mean count/grid cell	Complete	5.91	0.84	4	18	3499
2	Count/Survey	Complete	24.42	0.39	8	79	24,683
3a	Count/Survey	High climate scores	48.10	0.27	8	79	8152
3b	Count/Survey	Low climate scores	6.17	0.21	8	79	8526
3c	Count/Survey	High percent impervious	21.23	0.33	8	79	8148
3d	Count/Survey	Low percent impervious	31.16	0.38	8	79	8150

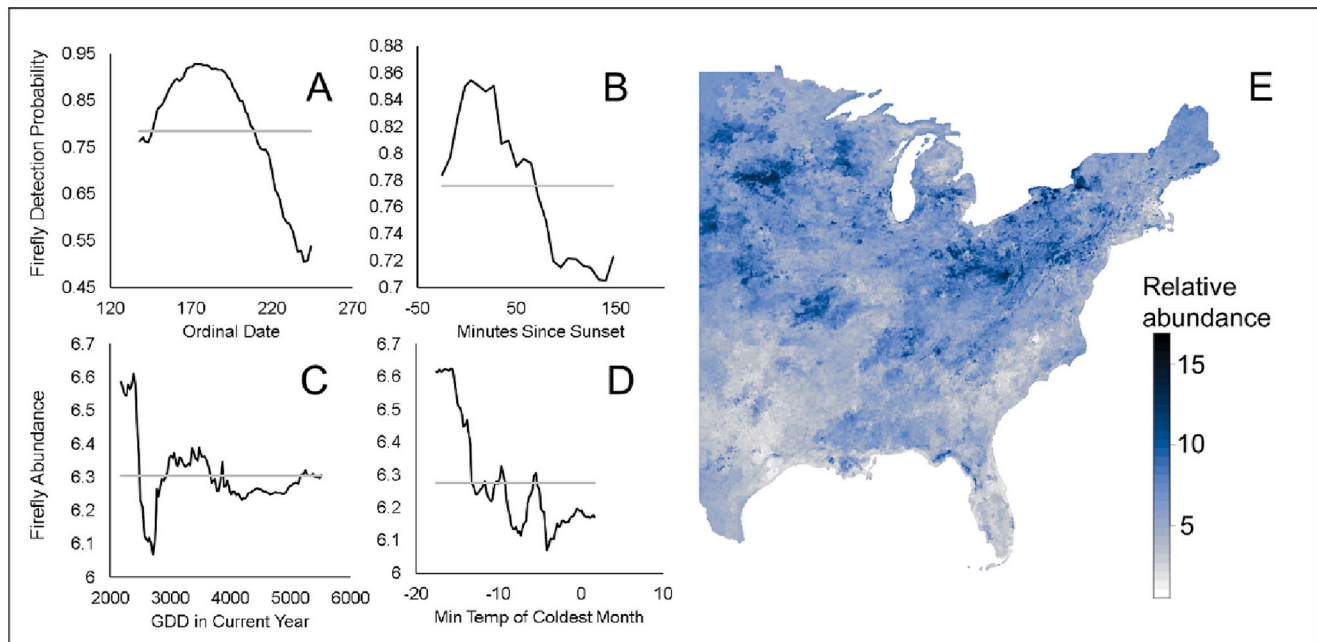


Fig. 2. Functional relationships from Tier 1 firefly random forest models. The probability of firefly detection (1/0) was driven by a suite of factors including minutes since sunset (A, B). Firefly counts were modeled using a variety of climate covariates including growing degree days (10th percentile; C, D). We also predicted the climate model across the eastern United States study area to visualize climate-driven patterns of abundance (E).

Table 2

The top ten variables for each ‘full’ random forest model (Tiers 2–3). Abbreviated variable names are followed by the variable importance score in parentheses. Weather variables are shown as either current year (“y_”), previous year (“p_”), or two years prior (“t_”). Shown variables include growing degree days (year to date; gdd_ytd), total growing degree days (gdd), frost free days (ffd), spring temperature minimum and maximum (sp_Tmin and Tmax), BioClim variables (bc05 = max temp of warmest month, bc06 = min temp in coldest month, bc09 = mean temp in driest quarter, bc19 = precipitation in coldest quarter), soil variables (percent sand [sand], and clay [clay], soil organic Carbon [soil_C], and available water storage [soil_wat]), and land cover variables (percent impervious [imperv], percent agriculture [ag], percent forest [forest], pesticide toxic load [pest] and night-time lights [NTL]).

Rank	All data	Low climate	High climate	High impervious	Low impervious
1	gdd_ytd (9.95)	gdd_ytd (1.62)	gdd_ytd (20.78)	gdd_ytd (7.37)	gdd_ytd (12.98)
2	p_gdd (4.23)	sand (0.72)	imperv (3.66)	clay (3.08)	p_gdd (4.58)
3	t_gdd (4.01)	p_gdd (0.65)	forest (3.15)	sand (2.66)	sand (4.35)
4	y_gdd (3.9)	pest (0.6)	NTL (2.51)	p_gdd (2.37)	t_gdd (4.23)
5	imperv (3.53)	forest (0.6)	ag (1.94)	t_gdd (1.88)	y_gdd (3.38)
6	sand (3.5)	p_bc05 (0.58)	y_bc19 (1.73)	y_gdd (1.87)	sp_Tmin (3.3)
7	sp_Tmin (2.76)	sp_Tmin (0.57)	y_gdd (1.7)	p_bc09 (1.75)	clay (3.28)
8	p_bc09 (2.64)	y_bc04 (0.55)	t_gdd (1.7)	p_bc06 (1.71)	p_bc05 (3.21)
9	NTL (2.47)	ag (0.54)	soil_C (1.65)	sp_Tmax (1.46)	sp_Tmax (2.88)
10	ag (2.45)	t_bc05 (0.52)	p_bc06 (1.61)	imperv (1.45)	y_ffd (2.79)

accumulated GDD at the time of survey, ii. previous year’s total GDD, and iii. Percent sand (Tables 1–2, see Table A.3 for results from all variables).

4. Discussion

Understanding factors associated with the health of insect populations has been established as a critical priority in global environmental conservation (Didham et al., 2020; Van Klink et al., 2020). Our analyses provide the most comprehensive assessment to date of factors driving population dynamics of adult North American fireflies. We found that variation in firefly abundance was explained by complex interactions among soil conditions, weather, and land cover characteristics – but even this pattern varied among landscapes with favorable/unfavorable climate and habitat conditions (Tables 1, 2), highlighting that abundance patterns depend on ecological context. In several important ways, our work provides empirical support for hypotheses proposed to explain potential factors limiting (or not limiting) North American firefly populations (i.e., habitat area, pesticides, light pollution, and climate; Lewis et al., 2020). We revealed support for habitat quality (including degradation via light pollution and loss via urbanization) and climate/weather as being primary factors limiting North American populations of bioluminescent fireflies. Notably, our results do not provide support for some previous hypotheses presented by Lewis et al. (2020): 1) climate and weather were *more* important than land-cover characteristics, and 2) we detected no negative impact of insecticide loads on firefly abundance. For the eastern U.S., our results provide much-needed descriptions of the factors influencing firefly abundance and suggest that actions that reduce, mitigate, or reverse habitat loss and climate change should have the strongest positive effects on future firefly populations (Fallon et al., 2019).

Our finding that firefly counts were strongly influenced by the time of day the observations were conducted, and accumulated GDDs at the time of the count make sense based on what is known about firefly life history (Lewis, 2016; Faust, 2017). Indeed, since the most common and widely distributed firefly species signal for mates around dusk (Lewis et al., 2021), surveys conducted earlier or later in the night (beyond the 30–60 min on either side of sunset) will be associated with reduced counts (Fig. 2). Interestingly, total GDDs from the previous one or two years were strong predictors of counts: these factors will be associated with variation in the population dynamics (either reproductive success of the adults in these years or the survival of the resulting immature

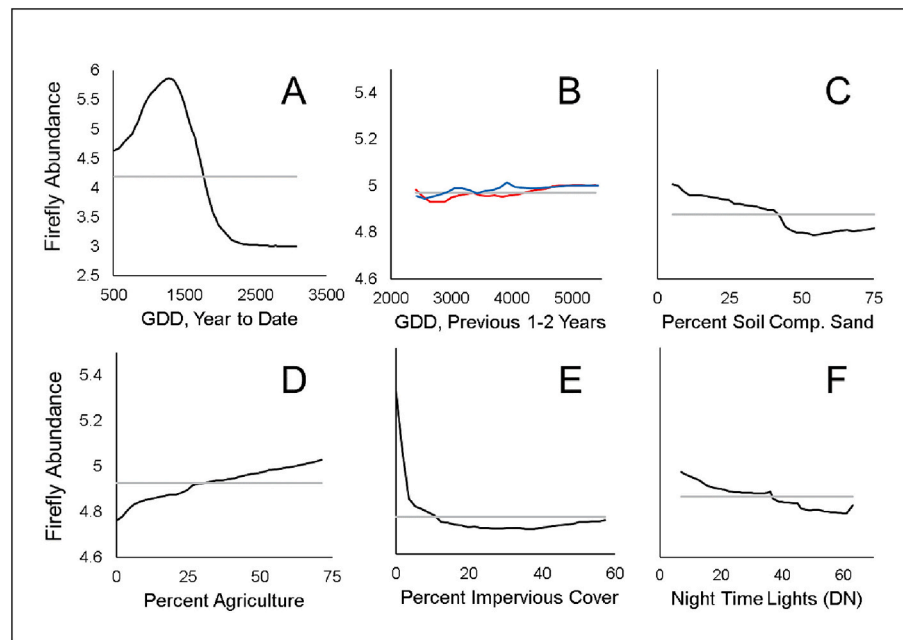


Fig. 3. Functional relationships from Tier 2 firefly random forest model, visualized using Partial Dependence Plots. X-axes are only shown from the 5th to the 95th percentiles to avoid making predictions for data-poor aspects of the environment. Shown are growing degree days accumulated at the time of survey (A), total growing degrees in the previous year (B, blue) and two years prior (B, red), percent soil composition sand (C), pesticide index (D), percent agriculture (D), Percent impervious cover (E), and night time lights “digital number” (DN; F).

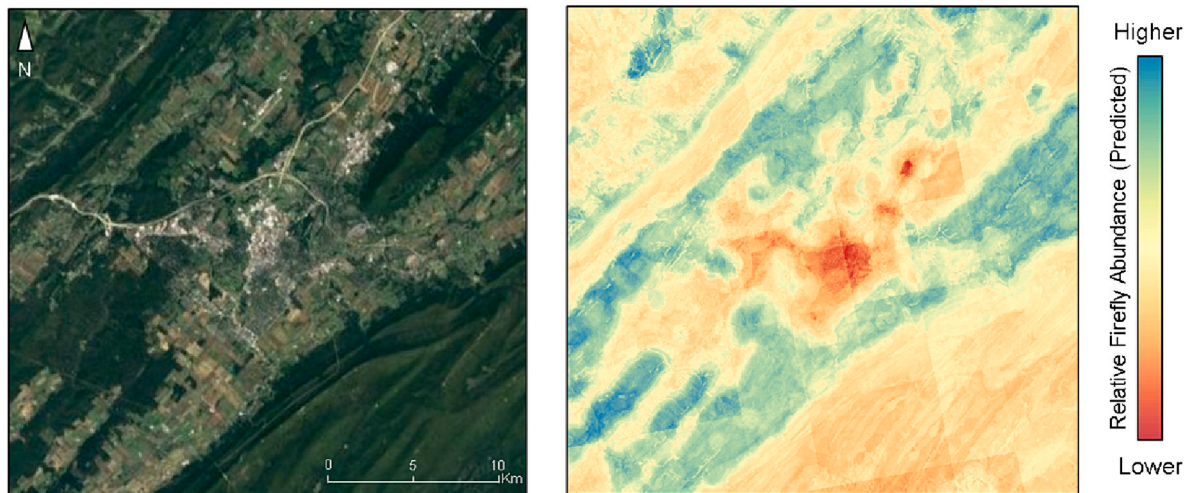


Fig. 4. A predictive map of firefly abundance incorporating detection probability, climate effects, weather effects, soils, and land cover features (including night-time lights and pesticides; left) for a small focal area in central Pennsylvania (Centre County). We predicted across this area because it has an extensive mix of urban, cropland, and natural cover (primarily deciduous forest; see corresponding aerial image, right).

stages; Evans et al., 2019). In the United States, there are >100 species of “flashing fireflies” that signal at night, and we assume that our analysis is likely skewed towards the most common and broadly distributed species that signal at dusk (e.g., *Photinus pyralis*) and where human observers are likely to be present such as residential yards. Future work to assess species/assemblages of conservation interest (e.g., *Phaenocarpa reticulata*) could still leverage citizen science in a more precise manner, as has also been done for some birds (e.g., Golden-winged Warbler Atlas Project [Rosenberg et al., 2016] and Nightjar Survey Network, [www.nightjars.org]).

Although previous work has suggested the importance of weather/climate in explaining aspects of firefly ecology (Evans et al., 2019; Harris et al., 2019), our work provides the first explicit assessment of the relative importance of climate in the context of other relevant predictors

like land cover, insecticides, and soils (Lewis et al., 2020). In a survey of firefly experts conducted in 2020, threats were ranked in the following order: habitat loss, light pollution, pesticide use, water pollution, drought, and higher temperatures (Lewis et al., 2020). In our study six of the top 10 variables within the ‘complete’ Tier 2 model (using all data and all variables) were related to climate/weather (Table 2). More specifically, our models suggested that an excess number of GDDs (Fig. 3) and warmer temperatures in winter (i.e., bc06, Fig. 2) are important for supporting bioluminescent fireflies across many portions of eastern North America (e.g., the Upper Midwest, Appalachian Mountains, etc.; Fig. 2; Evans et al., 2019). Notably, regions that had the highest firefly counts (Fig. 2E) also overlapped with US states that host high species richness of fireflies (e.g., parts of Florida, New York, Pennsylvania, see Fallon et al., 2019). In the short-term, firefly

abundance was positively associated with GDDs (Fig. 3B) meaning that warmer summers (and surveys where the preceding 1–2 summers were warmer) hosted higher firefly counts, on average. However, firefly abundance was negatively associated with metrics of temperature in both our Tier 1 climate model and Tier 2 complete model (frost free days, springtime temperatures, etc.) clearly indicates a pattern of fireflies being less abundant in warm regions of the eastern U.S. that accumulate many GDDs (e.g., portions of the Southeastern Plains of the U.S. [Ecoregion 8.3, sensu Omernik, 2004]). This paradoxical pattern may be because short-term warm periods support fireflies in their later life stages (e.g., activity of displaying adults) but may be otherwise negative for other life stages (e.g., immatures). Indeed, similar patterns have also been reported for other Coleoptera whereby lower depths and shorter durations of winter snow cover have been implicated as primary drivers of population declines (Harris et al., 2019; Homburg et al., 2019; Hallmann et al., 2020). Collectively, our results build upon a growing body of literature demonstrating the importance of climate/weather to fireflies, especially those metrics related to GDDs and temperature (Table 1; Lewis et al., 2020).

An important implication from our model results is that climate change is likely a serious threat to North American firefly populations (Lewis et al., 2020). Climatologists predict that climate change in eastern North America will yield warmer year-round temperatures (e.g., milder winters, hotter summers; Kukul and Irmak, 2018) and a higher frequency of extreme weather events like droughts and storms (Cook et al., 2018). Our climate and weather models suggested that fireflies were more sensitive to metrics of temperature than other weather/climate characteristics (e.g., precipitation) and our Tier 2 model demonstrated that temperature metrics were more predictive than any other ecological variables considered (i.e., landcover, soils, etc.). Even when separating high- and low-quality climatic regions in our Tier 3 models, GDDs in the previous two years was still in the top 10 predictors, and in low-quality climatic regions, springtime temperatures (both minimum and maximum) were top predictors. Temperature is known to be a strong predictor of phenological development of fireflies (Faust and Weston, 2009) and other insect taxa (Sambaraju et al., 2012; Purewaran et al., 2018), primarily due to its influence on eclosion and subsequent reproductive activities (e.g., courtship display; Lloyd, 1966; Lewis, 2016). Consequently, increased mean temperatures and other associated abiotic changes (e.g., humidity) predicted with ongoing climate change (Kukul and Irmak, 2018) are likely to hurt populations of fireflies. Although increased prevalence of extreme weather events like drought will likely influence all aspects of firefly life cycles (Evans et al., 2019).

Our results support prior studies suggesting that access to moist soils is essential for firefly populations to thrive, likely because such soils help immature stages avoid desiccation support soft-bodied invertebrate prey (McLean et al., 1972; Buschman, 1984; Viviani et al., 2012). Many species lay their eggs in moist soils near marshes, streams, and low-lying meadows (Barrows et al., 2008; Chatragadda, 2020). Evans et al. (2019) found soil water content was strongly associated with firefly abundance, but with a quadratic relationship, where low and high water content were associated with reduced numbers. Although soil water content was not within the top predictive variables in any of our data subsets (Table 2), firefly abundance declined with increased sand composition, likely because sandy soil is more likely to be well-drained (Huang and Hartemink, 2020) decreasing moisture for immature fireflies (Kakehashi et al., 2013). In many terrestrial soils, high sand composition is also associated with low nutrient levels (e.g., calcium; Oliveira et al., 2002; Huang and Hartemink, 2020), which would also hinder development of fireflies and their prey (Mänd et al., 2000; Lewis et al., 2020). Similarly, the negative relationship between firefly count and percent impervious cover is consistent with the fact that more impervious cover represents reductions in access to the soil surface where immature stages occur as well as vegetation for adult courtship (Lewis, 2016). Moreover, the pesticides applied to land cover with abundant impervious surfaces (i.e.,

non-agriculture) are not captured in the pesticide maps we used here (Douglas et al., 2020). For these reasons, some firefly ‘habitat certification’ programs penalize impervious cover in their assessments (Ridenhour, 2022). Our results agree with this general pattern, and indicate that even modest reductions in soil access (e.g., only 5–10 % impervious cover) can negatively influence firefly abundance (Fig. 2). Thus, our results collectively demonstrate that, within the best climatic conditions (e.g., ~2000–2500 GDDs), landscapes characterized by rich, moist soils and a low prevalence of impervious surfaces are most likely to host abundant firefly populations in the eastern United States.

One of the major threats to North American firefly populations is thought to be artificial light at night (e.g., street lights; Lewis et al., 2020). Anthropogenic lights are hypothesized to interfere with fireflies’ bioluminescent courtship displays (Costin and Boulton, 2016; Firebaugh and Haynes, 2016). Several small-scale studies (e.g., Hagen et al., 2015; Firebaugh and Haynes, 2016; Mbugua et al., 2020; Owens and Lewis, 2022) have demonstrated empirically and through observation that firefly abundance and reproductive success can be negatively affected by anthropogenic lights, and our results support the hypothesis that anthropogenic light sources may harm firefly populations. One major caveat of this finding, however, is that the effects of anthropogenic light are often confounded by those of urbanization (i.e., buildings and structures that produce artificial lights at night), which are correlated and both yield a negative effect (Hagen et al., 2015; Fenoglio et al., 2021). Thus, it remains difficult to disentangle the relative effects at broad scales of urbanization and anthropogenic light per se on firefly abundance (Picchi et al., 2013), an issue also present within other relevant datasets (e.g., pesticide toxic load with correlated effects of modern agricultural practices; Douglas et al., 2020, 2022). Regardless, highly urbanized places dominated by impervious cover and bright lights at night are likely to host fewer fireflies than similar places that are dark at night and allow access to soil (e.g., natural cover types and agriculture, etc.; Fig. 4).

Of the firefly-land-cover relationships revealed by our analyses, some were very intuitive while others were surprising. Our finding that man-made impervious cover was a predictor of firefly habitat quality makes intuitive sense because many fireflies evolved in systems that largely lack this cover type (Faust, 2010; Heckscher, 2010, 2021). Natural/semi-natural ecosystems provide herbaceous and woody cover for adults to court and oviposit while the understory and forest floor provide essential habitat for immature fireflies (Buschman, 1984; De Cock et al., 2014; Faust et al., 2019). With this pattern in mind, we were surprised to find firefly abundance was positively associated with percent agricultural cover (Fig. 2). This is surprising because agricultural areas, like those we found to have high firefly densities, also have the highest pesticide toxic loads (Douglas et al., 2020). For many insect species, increased area of agricultural cover is associated with declines in habitat quality and heightened risk of population loss (Outhwaite et al., 2022; Ziesche et al., 2023). It is possible that, for some species of fireflies, agricultural areas promote the structural requirements for adult courtship (e.g., relatively open “meadow-like” conditions), especially within pastures and fallow fields (Lewis et al., 2020; Fallon et al., 2021) while also providing the moist, nutrient-rich soils that support larval prey items. In fact, other efforts have demonstrated that insect communities can thrive in agricultural landscapes, especially when conservation practices are applied to enhance ecosystem function (Hirsh et al., 2013; Schulte et al., 2017). It is also possible that, compared to suburban or forested settings among others, agricultural areas offer clearer lines of sight that could facilitate observing more fireflies in each observation because for May, June and some of July, crops remain relatively short. Future work assessing the presence and importance of such a bias (e.g., using distance sampling; Buckland et al., 2015) would be valuable. Lastly, although agricultural areas are presumed to have higher pesticide loads than adjacent natural/semi-natural habitats (Douglas et al., 2020), the relative robustness of some fireflies (e.g., *Photuris versicolor*, *Photinus pyralis*) to neonicotinoid insecticides may allow them to persist

in habitats that would be of low-quality to other insects (Pearsons et al., 2021). It is important to recognize that the Firefly Watch dataset does not discern individual species, the conservation status of which vary widely (Fallon et al., 2021). Given the strong possibility that agricultural habitats are dominated by a few very common species (like *Photinus pyralis*) that are not imperiled, our finding that firefly abundance was positively associated with agricultural cover should be interpreted with care.

5. Conclusion

Through use of a large citizen-science dataset, our results suggest that climate change might be a greater threat to North American firefly populations than previously appreciated. Still, other factors (especially land cover characteristics and landscape context) remain strong predictors of firefly abundance in the eastern United States. Although we detected a strong negative effect of urbanization on firefly abundance (i. e., percent impervious cover), fireflies from the Firefly Watch dataset were common in many suburban habitats and agricultural lands (Fig. 4). This finding points to a potential opportunity to develop firefly-friendly conservation practices, especially those that maintain/enhance access to the soil surface. Given that many firefly species are habitat specialists (Reed et al., 2020), species-specific monitoring efforts (either via citizen science or biologist-led censuses) would help to disentangle the habitat patterns revealed by our analyses and assess the extent to which such relationships hold true for individual species. In sum, this study demonstrates that effective conservation of North American firefly populations will likely depend upon 1) continued and enhanced monitoring via programs like Firefly Watch, 2) efforts to mitigate impacts of climate change, and 3) implementation of insect-friendly conservation practices within potential habitats (Dickinson et al., 2010; Mesaglio et al., 2021).

CRedit authorship contribution statement

Darin J. McNeil: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Melanie Kammerer:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sarah E. Lower:** Writing – review & editing, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation. **John F. Tooker:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Christina M. Grozinger:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study are publicly available and noted as such.

Acknowledgment

We greatly appreciate all the citizen scientists that contributed data to Firefly Watch as well as the Firefly Watch team for coordinating, administering, and advertising the program. We are grateful to Don Salvatore for his contributions to completing this manuscript. Funding for this project was provided by the Penn State Insect Biodiversity Center, the Huck Institutes of the Life Sciences, and the College of

Agricultural Sciences via the National Institute of Food and Agriculture and Hatch Appropriations under Project #PEN04606 and Accession #1009362. The funders of this work had no involvement in collection, analysis or interpretation of data, in the writing of this manuscript, or the decision to submit this article for publication. This project was supported by McIntire-Stennis Capacity Grant #KY009043. Finally, we are very grateful to one anonymous reviewer and D. Wagner for both providing excellent comments that improved this manuscript substantially.

Table A.1

Independent variables included in random forest analyses of firefly abundance. Shown for each variable are the mean, standard deviation (SD), minimum (Min), and maximum (Max). For each, we also include units, time period (where applicable; current year, previous year, or two years prior), and BioClim number (bc##), where applicable.

Variable name	Mean	SD	Min	Max
Climate Score	5	4	0	20
% Soil Percent Clay	21	10	0	53
% Soil Percent Earth	77	20	0	100
% Soil Percent Sand	37	21	1	99
Aspect (East-West)	−0.01	0.73	−1	1
Slope (Degrees)	3	3	0	44
Soil Available Water Storage	119	51	0	437
Soil Bulk Density	1424	76	633	1639
Soil Crop Index	0.49	0.23	0.01	0.98
Soil Organic Carbon	7495	8100	37	114,315
Soil Root Zone Depth	124	39	0	150
% Barren Cover	0.2	1.2	0	29
% Forest Cover	24	23	0	100
% Grass/pasture Cover	8	13	0	93
% Impervious Cover	18	17	0	89
% Open Water Cover	1	5	0	75
% Rowcrops Cover	8	15	0	100
% Shrubland Cover	1	4	0	87
% Wetland Cover	4	10	0	99
% Agriculture	16	22	0	100
Night Time Lights	44	20	0	63
Pesticide Load (LD ₅₀ /ha)	0.33	0.74	0	19
# Accumulated GDD (At Survey)	1521	793	0	7288
# Frost-free Days (Prev Year)	232	36	122	366
# Frost-free Days (Survey Year)	235	37	101	366
# Frost-free Days (Two Years)	231	36	122	366
# GDD (Prev Year)	3480	874	1556	7792
# GDD (Survey Year)	3510	885	1412	7610
# GDD (Two Years)	3464	869	1582	7522
Isothermality (Prev Year; C; “bc03”)	0.29	0.04	0.17	0.51
Isothermality (Survey Year; C; “bc03”)	0.29	0.04	0.17	0.53
Isothermality (Two Years; C; “bc03”)	0.3	0.04	0.17	0.53
Longest Precip-free Gap (Prev Year; Days)	5	1	3	18
Longest Precip-free Gap (Survey Year; Days)	5	1	3	19
Longest Precip-free Gap (Two Years; Days)	6	1	3	23
Max Temp of Warmest Month (Prev Year; C; “bc05”)	30	3	22	40
Max Temp of Warmest Month (Survey Year; C; “bc05”)	30	3	22	41
Max Temp of Warmest Month (Two Years; C; “bc05”)	30	3	21	41
Mean Diurnal Range (Prev Year; C; “bc02”)	11	1	6	17
Mean Diurnal Range (Survey Year; C; “bc02”)	11	1	6	16
Mean Diurnal Range (Two Years; C; “bc02”)	11	1	6	16
Mean Temp of Driest Quarter (Prev Year; C; “bc09”)	9	10	−14	32
Mean Temp of Driest Quarter (Survey Year; C; “bc09”)	9	9	−14	32
Mean Temp of Driest Quarter (Two Years; C; “bc09”)	10	10	−13	32
Mean Temp of Wettest Quarter (Prev Year; C; “bc08”)	17	7	−5	30
Mean Temp of Wettest Quarter (Survey Year; C; “bc08”)	17	7	−5	30
Mean Temp of Wettest Quarter (Two Years; C; “bc08”)	17	7	−4	30

(continued on next page)

Table A.1 (continued)

Variable name	Mean	SD	Min	Max
Min Temp of Coldest Month (Prev Year; C; "bc06")	−7	5	−27	15
Min Temp of Coldest Month (Survey Year; C; "bc06")	−7	5	−24	16
Min Temp of Coldest Month (Two Years; C; "bc06")	−7	5	−28	18
Precip of Coldest Quarter (Prev Year; cm; "bc19")	239	89	14	705
Precip of Coldest Quarter (Survey Year; cm; "bc19")	234	93	20	701
Precip of Coldest Quarter (Two Years; cm; "bc19")	233	86	14	784
Precip of Driest Quarter (Prev Year; cm; "bc17")	167	58	2	416
Precip of Driest Quarter (Survey Year; cm; "bc17")	162	57	3	377
Precip of Driest Quarter (Two Years; cm; "bc17")	161	54	0	437
Precip of Warmest Quarter (Prev Year; cm; "bc18")	325	117	12	954
Precip of Warmest Quarter (Survey Year; cm; "bc18")	339	118	19	994
Precip of Warmest Quarter (Two Years; cm; "bc18")	322	113	22	1084
Precip of Wettest Month (Prev Year; cm; "bc13")	211	66	72	682
Precip of Wettest Month (Survey Year; cm; "bc13")	209	64	79	705
Precip of Wettest Month (Two Years; cm; "bc13")	205	65	61	602
Precip Seasonality (Prev Year; CoV; "bc15")	0.56	0.15	0.18	1.41
Precip Seasonality (Survey Year; CoV; "bc15")	0.56	0.15	0.18	1.63
Precip Seasonality (Two Years; CoV; "bc15")	0.57	0.15	0.19	1.62
Spring Max Temp (Celsius)	18	4	7	31
Spring Min Temp (Celsius)	6	4	−6	22
Spring Precip (cm)	300	98	22	1158
Temp Annual Range (Prev Year; C; "bc07")	37	4	16	53
Temp Annual Range (Survey Year; C; "bc07")	37	3	15	50
Temp Annual Range (Two Years; C; "bc07")	37	3	15	51
Temp Seasonality (Prev Year; C; "bc04")	9	1	3	14
Temp Seasonality (Survey Year; C; "bc04")	9	1	3	14
Temp Seasonality (Two Years; C; "bc04")	9	1	3	14
Total Precip (Prev Year; cm)	1174	247	348	2462
Total Precip (Survey Year; cm)	1165	242	309	2687
Total Precip (Two Years; cm)	1135	243	306	2444
Winter Max Temp (Celsius)	5	5	−10	26
Winter Min Temp (Celsius)	−4	4	−22	18
Winter Precip (cm)	248	97	17	872

Table A.2

Results for the detection and climate models (Tier 1). Shown are the names for each variable within their respective models as well as the Variable Importance Scores (VIS) for each.

Variable name	VIS
Tier 1 - Detection Model	
Ordinal Date	0.04
Minutes Since Sunrise	0.02
Temperature	0.01
Precipitation Index	0.00
Cloud Cover Index	0.00
Windspeed Index	0.00
Moonlight Index	0.00
Tier 1 - Climate Model	
Accumulated GDD (10th Percentile)	8.13
Min Temp of Coldest Month (10th Percentile)	7.77
# Frost-free Days (10th Percentile)	7.61
Temp Seasonality (90th Percentile)	6.22
Temp Annual Range (90th Percentile)	6.14
Max Temp of Warmest Month (90th Percentile)	5.36

(continued on next column)

Table A.2 (continued)

Variable name	VIS
Longest Precip-free Gap (90th Percentile)	5.36
Precip of Coldest Quarter (10th Percentile)	4.39
Isothermality (90th Percentile)	4.54
Precip Seasonality (90th Percentile)	4.37
Mean Temp of Driest Quarter (90th Percentile)	4.45
Mean Temp of Wettest Quarter (10th Percentile)	4.40
Total Precip (10th Percentile)	3.78
Precip of Wettest Month (90th Percentile)	3.40
Precip of Driest Quarter (10th Percentile)	3.12
Mean Diurnal Range (90th Percentile)	2.99
Precip of Warmest Quarter (90th Percentile)	2.40

Table A.3

Variable Importance Scores from each full-variable analysis conducted using the Firefly Watch dataset: High climate score subset ("High Clim."), low climate score subset ("Low Clim."), high percent impervious subset ("High Imp."), low percent impervious subset ("Low Imp."), and the full dataset.

Variable name	Full dataset	Low Clim.	High Clim.	High Imp.	Low Imp.
Aggregated Climate Score	18.66	0.93	18.92	11.73	21.88
Soil and topography variables					
% Soil Percent Clay	1.69	0.42	1.55	0.94	1.56
% Soil Percent Earth	1.30	0.48	1.3	1.17	1.12
% Soil Percent Sand	2.43	0.39	1.32	3.08	3.28
Aspect (East-West)	3.50	0.72	1.09	2.66	4.35
Slope (Degrees)	1.22	0.50	1.65	0.73	1.62
Soil Available Water	1.59	0.47	1.22	0.92	2.42
Storage					
Soil Bulk Density	0.73	0.22	1.10	0.53	0.50
Soil Crop Index	0.44	0.13	0.46	0.58	0.50
Soil Organic Carbon	1.10	0.49	0.99	1.00	1.26
Soil Root Zone Depth	0.99	0.32	1.03	1.36	0.83
Landcover characteristics					
Night Time Lights	2.47	0.28	2.51	0.45	1.14
Pesticide Load	1.75	0.60	1.32	0.84	1.23
% Open Water Cover	0.14	0.08	0.19	0.11	0.30
% Barren Cover	0.06	0.12	0.06	0.03	0.04
% Shrubland Cover	0.25	0.02	0.23	0.02	0.45
% Grass/pasture Cover	1.43	0.48	0.79	0.14	1.22
% Rowcrops Cover	1.50	0.36	1.21	0.33	1.48
% Forest Cover	2.00	0.60	3.15	0.74	1.22
% Wetland Cover	0.43	0.20	0.69	0.19	0.81
% Impervious Cover	3.53	0.47	3.66	1.45	2.23
% Agriculture Cover	2.45	0.54	1.94	0.37	2.52
Climate and weather variables					
# Accumulated GDD (At Survey)	9.95	1.62	20.78	7.37	12.98
# Frost-free Days (Prev Year)	1.22	0.21	0.93	0.79	1.86
# Frost-free Days (Survey Year)	1.55	0.19	1.08	0.90	2.79
# Frost-free Days (Two Years)	2.15	0.19	1.09	0.97	1.87
# GDD (Prev Year)	4.23	0.65	1.29	2.37	4.58
# GDD (Survey Year)	3.90	0.52	1.70	1.87	3.38
# GDD (Two Years)	4.01	0.47	1.70	1.88	4.23
Isothermality (Prev Year; C)	1.12	0.32	1.08	0.94	0.94
Isothermality (Survey Year; C)	1.07	0.33	1.01	0.76	1.03
Isothermality (Two Years; C)	1.08	0.33	1.03	0.88	1.07
Longest Precip-free Gap (Prev Year; Days)	1.03	0.16	0.94	0.67	1.58
Longest Precip-free Gap (Survey Year; Days)	0.99	0.20	0.89	0.64	1.14

(continued on next page)

Table A.3 (continued)

Variable name	Full dataset	Low Clim.	High Clim.	High Imp.	Low Imp.
Longest Precip-free Gap (Two Years; Days)	0.89	0.23	0.86	0.66	0.97
Max Temp of Warmest Month (Prev Year; C)	2.38	0.58	1.19	1.14	3.21
Max Temp of Warmest Month (Survey Year; C)	1.55	0.38	1.07	0.90	2.30
Max Temp of Warmest Month (Two Years; C)	1.93	0.52	1.29	0.90	1.79
Mean Diurnal Range (Prev Year; C)	0.94	0.15	0.90	0.78	0.91
Mean Diurnal Range (Survey Year; C)	0.98	0.24	0.98	0.82	1.02
Mean Diurnal Range (Two Years; C)	0.95	0.23	1.00	0.90	0.96
Mean Temp of Driest Quarter (Prev Year; C)	2.64	0.47	1.12	1.75	2.31
Mean Temp of Driest Quarter (Survey Year; C)	1.40	0.51	0.90	1.33	1.13
Mean Temp of Driest Quarter (Two Years; C)	1.10	0.25	0.86	0.75	1.08
Mean Temp of Wettest Quarter (Celcius)	1.03	0.35	1.02	1.09	0.93
Mean Temp of Wettest Quarter (Celcius)	0.98	0.22	0.82	0.82	0.93
Mean Temp of Wettest Quarter (Celcius)	0.82	0.18	0.76	0.54	1.01
Min Temp of Coldest Month (Prev Year; C)	1.50	0.43	1.61	1.71	1.39
Min Temp of Coldest Month (Survey Year; C)	1.75	0.38	1.09	1.07	1.94
Min Temp of Coldest Month (Two Years; C)	1.43	0.40	1.28	1.31	1.27
Precip of Coldest Quarter (Prev Year; cm)	1.00	0.18	1.05	0.74	0.96
Precip of Coldest Quarter (Survey Year; cm)	0.94	0.21	1.73	0.88	0.80
Precip of Coldest Quarter (Two Years; cm)	0.73	0.16	0.98	0.70	0.77
Precip of Driest Quarter (Prev Year; cm)	0.82	0.17	0.97	0.62	0.92
Precip of Driest Quarter (Survey Year; cm)	0.77	0.18	1.01	0.68	0.76
Precip of Driest Quarter (Two Years; cm)	0.69	0.13	0.89	0.61	0.89
Precip of Warmest Quarter (Prev Year; cm)	0.83	0.19	1.54	0.79	0.93
Precip of Warmest Quarter (Survey Year; cm)	0.77	0.19	0.80	0.67	0.83
Precip of Warmest Quarter (Two Years; cm)	0.71	0.20	0.81	0.56	0.75
Precip of Wettest Month (Prev Year; cm)	0.66	0.27	1.14	0.59	0.85
Precip of Wettest Month (Survey Year; cm)	0.75	0.12	1.21	0.51	1.21
Precip of Wettest Month (Two Years; cm)	0.94	0.14	0.77	0.58	1.04
Precip Seasonality (Prev Year; CoV)	0.81	0.43	0.86	0.65	1.33
Precip Seasonality (Survey Year; CoV)	0.90	0.16	0.96	0.68	1.30
Precip Seasonality (Two Years; CoV)	0.98	0.36	0.78	0.60	1.21
Spring Max Temp (Celcius)	2.00	0.46	1.36	1.46	2.88
Spring Min Temp (Celcius)	2.76	0.57	1.49	1.42	3.30
Spring Precip (cm)	0.75	0.20	0.94	0.62	0.83
Temp Annual Range (Prev Year; C)	0.89	0.32	0.98	1.06	0.84
Temp Annual Range (Survey Year; C)	0.80	0.23	0.76	0.73	0.78
Temp Annual Range (Two Years; C)	0.94	0.22	1.16	0.91	0.89
Temp Seasonality (Prev Year; C)	1.15	0.29	1.40	1.27	0.96

(continued on next column)

Table A.3 (continued)

Variable name	Full dataset	Low Clim.	High Clim.	High Imp.	Low Imp.
Temp Seasonality (Survey Year; C)	1.00	0.55	1.08	1.01	1.15
Temp Seasonality (Two Years; C)	0.98	0.30	0.91	0.90	1.07
Total Precip (Prev Year; cm)	0.86	0.22	1.52	0.87	0.81
Total Precip (Survey Year; cm)	0.77	0.18	1.10	0.60	1.02
Total Precip (Two Years; cm)	0.78	0.12	1.01	0.79	0.78
Winter Max Temp (Celcius)	2.11	0.44	1.29	1.25	2.03
Winter Min Temp (Celcius)	2.05	0.33	1.12	1.21	2.00
Winter Precip (cm)	1.20	0.16	0.93	1.04	0.92

References

- Barrows, E.M., Arsenault, S.B., Grenier, N.P., 2008. Firefly (Coleoptera: Lampyridae) flight periods, sex ratios, and habitat frequencies in a United States mid-Atlantic freshwater tidal marsh, low forest, and their ecotone. *Banisteria* 31, 47–52.
- Bradter, U., Mair, L., Jönsson, M., Knappe, J., Singer, A., Snäll, T., 2018. Can opportunistically collected citizen science data fill a data gap for habitat suitability models of less common species? *Methods Ecol. Evol.* 9, 1667–1678.
- Brooks, D.R., Bader, J.E., Clark, S.J., Monteith, D.T., Andrews, C., Corbett, S.J., Beaumont, D.A., Chapman, J.W., 2012. Large carabid beetle declines in a United Kingdom monitoring network increases evidence for a widespread loss in insect biodiversity. *J. Appl. Ecol.* 49, 1009–1019.
- Brooks, T.M., Mittermeier, R.A., Mittermeier, C.G., Da Fonseca, G.A., Rylands, A.B., Konstant, W.R., Flick, P., Pilgrim, J., Oldfield, S., Magin, G., Hilton-Taylor, C., 2002. Habitat loss and extinction in the hotspots of biodiversity. *Conserv. Biol.* 16, 909–923.
- Brown, K.S., 1997. Diversity, disturbance, and sustainable use of Neotropical forests: insects as indicators for conservation monitoring. *J. Insect Conserv.* 1, 25–42.
- Buckland, S.T., Rexstad, E.A., Marques, T.A., Oedekoven, C.S., 2015. *Distance Sampling: Methods and Applications*. Springer, New York, USA.
- Buschman, L.L., 1984. Biology of the firefly *Pyroctomena lucifera* (Coleoptera: Lampyridae). *Fla. Entomol.* 67, 529–542.
- Calovi, M., Grozinger, C.M., Miller, D.A., Goslee, S.C., 2021. Summer weather conditions influence winter survival of honey bees (*Apis mellifera*) in the northeastern United States. *Sci. Rep.* 11, 1553.
- Cardoso, P., Erwin, T.L., Borges, P.A., New, T.R., 2011. The seven impediments in invertebrate conservation and how to overcome them. *Biol. Conserv.* 144, 2647–2655.
- Chatragadda, R., 2020. Decline of luminous firefly *Abcondita chinensis* population in Barrankula, Andhra Pradesh, India. *Int. J. Trop. Insect Sci.* 40, 461–465.
- Cook, B.L., Mankin, J.S., Anchukaitis, K.J., 2018. Climate change and drought: from past to future. *Curr. Clim. Chang. Rep.* 4, 164–179.
- Costin, K.J., Boulton, A.M., 2016. A field experiment on the effect of introduced light pollution on fireflies (Coleoptera: Lampyridae) in the Piedmont region of Maryland. *Coleopt. Bull.* 70, 84–86.
- Crossley, M.S., Meier, A.R., Baldwin, E.M., Berry, L.L., Crenshaw, L.C., Hartman, G.L., Lagos-Kutz, D., Nichols, D.H., Patel, K., Varriano, S., Snyder, W.E., 2020. No net insect abundance and diversity declines across US long term ecological research sites. *Nature Ecol. Evol.* 4, 1368–1376.
- De Cock, R., Faust, L., Lewis, S., 2014. Courtship and mating in *Phausis reticulata* (Coleoptera: Lampyridae): male flight behaviors, female glow displays, and male attraction to light traps. *Fla. Entomol.* 97, 1290–1307.
- Dickinson, J.L., Zuckerberg, B., Bonter, D.N., 2010. Citizen science as an ecological research tool: challenges and benefits. *Ann. Rev. Ecol. Syst.* 41, 149–172.
- Didham, R.K., Basset, Y., Collins, C.M., Leather, S.M., Littlewood, N.A., Menz, M.H.M., Muller, J., Packer, L., Saunders, M.E., Schonrogge, K., Stewart, A.J.A., Yanoviak, S. P., Hassall, C., 2020. Interpreting insect declines: seven challenges and a way forward. *Insect Conserv. Divers.* 13, 103–114.
- Dirzo, R., Young, H.S., Galetti, M., Ceballos, G., Isaac, N.J., Collen, B., 2014. Defaunation in the Anthropocene. *Science* 345, 401–406.
- Douglas, M.R., Sponsler, D.B., Lonsdorf, E.V., Grozinger, C.M., 2020. County-level analysis reveals a rapidly shifting landscape of insecticide hazard to honey bees (*Apis mellifera*) on US farmland. *Sci. Rep.* 10, 1–11.
- Douglas, M.R., Baisley, P., Soba, S., Kammerer, M., Lonsdorf, E.V., Grozinger, C.M., 2022. Putting pesticides on the map for pollinator research and conservation. *Sci. Data* 9, 571.
- Evans, T.R., Salvatore, D., van de Pol, M., Musters, C.J.M., 2019. Adult firefly abundance is linked to weather during the larval stage in the previous year. *Ecol. Entomol.* 44, 265–273.

- Fallon, C., Hoyle, S., Lewis, S., Owens, A., Lee-Mäder, E., Black, S.H., Jepsen, S., 2019. Conserving the Jewels of the Night: Guidelines for Protecting Fireflies in the United States and Canada. The Xerces Society for Invertebrate Conservation, Portland, OR.
- Fallon, C.E., Walker, A.C., Lewis, S., Cicero, J., Faust, L., Heckscher, C.M., Pérez-Hernández, C.X., Pfeiffer, B., Jepsen, S., 2021. Evaluating firefly extinction risk: initial red list assessments for North America. *PLoS One* 16 (11), e0259379.
- Faust, L.F., 2010. Natural history and flash repertoire of the synchronous firefly *Photinus carolinus* (Coleoptera: Lampyridae) in the Great Smoky Mountains National Park. *Fla. Entomol.* 93, 208–217.
- Faust, L.F., 2012. Fireflies in the snow: observations on two early-season arboreal fireflies *Ellychnia corrusca* and *Pyroctomena borealis*. *Lampyrid* 2, 48–71.
- Faust, L.F., 2017. Fireflies, glow-worms, and lightning bugs: identification and natural history of the fireflies of the eastern and central United States and Canada. University of Georgia Press, Athens, Georgia, USA.
- Faust, L.F., Weston, P.A., 2009. Degree-day prediction of adult emergence of *Photinus carolinus* (Coleoptera: Lampyridae). *Environ. Entomol.* 38, 1505–1512.
- Faust, L.F., Hughes, L.S., Zloba, M.H., Farrington, H.L., 2019. Life history and updated range extension of *Photinus scintillans* (Coleoptera: Lampyridae) with new Ohio records and regional observations for several firefly species. *Ohio Biological Survey Notes* 9, 16–34.
- Fenoglio, M.S., Calviño, A., González, E., Salvo, A., Videla, M., 2021. Urbanisation drivers and underlying mechanisms of terrestrial insect diversity loss in cities. *Ecol. Entomol.* 46, 757–771.
- Fick, S.E., Hijmans, R.J., 2017. WorldClim 2: new 1km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* 37, 4302–4315.
- Filazzola, A., Matter, S.F., MacIvor, J.S., 2021. The direct and indirect effects of extreme climate events on insects. *Sci. Total Environ.* 769, 145161.
- Fink, D., Auer, T., Johnston, A., Strimas-Mackey, M., Robinson, O., Ligocki, S., Hochachka, W., Wood, C., Davies, I., Iliff, M., Seitz, L., 2020. eBird Status and Trends, Data Version: 2019; Released: 2020. Cornell Lab of Ornithology, Ithaca, New York, USA.
- Firebaugh, A., Haynes, K.J., 2016. Experimental tests of light-pollution impacts on nocturnal insect courtship and dispersal. *Oecologia* 182 (4), 1203–1211.
- Forister, M.L., Pelton, E.M., Black, S.H., 2019. Declines in insect abundance and diversity: we know enough to act now. *Conserv. Sci. Pract.* 1 (8), e80.
- Forister, M.L., Halsch, C.A., Nice, C.C., Fordyce, J.A., Diltz, T.E., Oliver, J.C., Prudic, K.L., Shapiro, A.M., Wilson, J.K., Glassberg, J., 2021. Fewer butterflies seen by community scientists across the warming and drying landscapes of the American west. *Science* 371, 1042–1045.
- Greenwell, B., 2018. Pdp: partial dependence plots. In: R Package. Version 0.7.0.
- Hagen, O., Santos, R.M., Schlindwein, M.N., Viviani, V.R., 2015. Artificial night lighting reduces firefly (Coleoptera: Lampyridae) occurrence in Sorocaba. *Brazil. Adv. Entomol.* 3, 24.
- Hallmann, C.A., Sorg, M., Jongejans, E., Siepel, H., Hofland, N., Schwan, H., Stenmans, W., Müller, A., Sumser, H., Hörrn, T., Goulson, D., de Kroon, H., 2017. More than 75 percent decline over 27 years in total flying insect biomass in protected areas. *PLoS One* 12, e0185809.
- Hallmann, C.A., Zeegers, T., van Klink, R., Vermeulen, R., van Wielink, P., Spijkers, H., van Deijk, J., van Steenis, W., Jongejans, E., 2020. Declining abundance of beetles, moths and caddisflies in the Netherlands. *Insect Conserv. Divers.* 13 (2), 127–139.
- Halsch, C.A., Shapiro, A.M., Fordyce, J.A., Nice, C.C., Thorne, J.H., Waetjen, D.P., Forister, M.L., 2021. Insects and recent climate change. *Proc. Natl. Acad. Sci.* 118, e2002543117.
- Harris, J.E., Rodenhouse, N.L., Holmes, R.T., 2019. Decline in beetle abundance and diversity in an intact temperate forest linked to climate warming. *Biol. Conserv.* 240, 108219.
- Harvey, J.A., Heinen, R., Armbrrecht, I., Basset, Y., Baxter-Gilbert, J.H., Bezemer, T.M., de Kroon, H., 2020. International scientists formulate a roadmap for insect conservation and recovery. *Nature Ecol. Evol.* 4, 174–176.
- Harvey, J.A., Tougeron, K., Gols, R., Heinen, R., Abarca, M., Abram, P.K., Chown, S.L., 2023. Scientists' warning on climate change and insects. *Ecol. Monogr.* 93, e1553.
- Heckscher, C.M., 2010. Delaware *Photuris* fireflies (Coleoptera: Lampyridae): New state records, conservation status, and habitat associations. *Entomol. News* 121, 498–505.
- Heckscher, C.M., 2021. Four new species of north American fireflies from isolated peatlands with reference to species determination of *Photuris* Dejean (Coleoptera: Lampyridae). *Northeast. Nat.* 28, 277–295.
- Hirsh, S.M., Mabry, C.M., Schulte, L.A., Liebman, M., 2013. Diversifying agricultural catchments by incorporating tallgrass prairie buffer strips. *Ecol. Restor.* 31 (2), 201–211.
- Homburg, K., Drees, C., Boutaud, E., Nolte, D., Schuett, W., Zumstein, P., von Ruschkowski, P., Assmann, T., 2019. Where have all the beetles gone? Long-term study reveals carabid species decline in a nature reserve in northern Germany. *Insect Conserv. Divers.* 12, 268–277.
- Høye, T.T., Post, E., Schmidt, N.M., Trøjsgaard, K., Forchhammer, M.C., 2013. Shorter flowering seasons and declining abundance of flower visitors in a warmer Arctic. *Nat. Clim. Chang.* 3, 759–763.
- Huang, J., Hartemink, A.E., 2020. Soil and environmental issues in sandy soils. *Earth Sci. Rev.* 208, 103295.
- Janzen, D.H., Hallwachs, W., 2019. Perspective: where might be many tropical insects? *Biol. Conserv.* 233, 102–108.
- Kakehashi, K., Kuranishi, R.B., Kamata, N., 2013. Environmental factors affecting the spatial distribution and activity of firefly larvae *Luciola parvula* (Coleoptera: Lampyridae: Luciolinae): high activity under rich soil moisture. *Japan. J. Conserv. Ecol.* 18, 45–54.
- Kammerer, M., Goslee, S.C., Douglas, M.R., Tooker, J.F., Grozinger, C.M., 2021. Wild bees as winners and losers: relative impacts of landscape composition, quality, and climate. *Glob. Chang. Biol.* 27, 1250–1265.
- Kevan, P.G., Viana, B.F., 2003. The global decline of pollination services. *Biodiversity* 4, 3–8.
- Kukal, M.S., Irmak, S., 2018. US agro-climate in 20 the century: growing degree days, first and last frost, growing season length, and impacts on crop yields. *Sci. Rep.* 8, 1–14.
- Lewis, S.M., 2016. Silent Sparks: The Wondrous World of Fireflies. Princeton University Press, Princeton, New Jersey, USA.
- Lewis, S.M., Wong, C.H., Owens, A., Fallon, C., Jepsen, S., Thancharoen, A., Chiahung, W., de Cock, R., Novak, M., Lopez-Palafox, T., Khoo, V., Reed, J.M., 2020. A global perspective on firefly extinction threats. *BioScience* 70, 157–167.
- Lewis, S.M., Thancharoen, A., Wong, C.H., López-Palafox, T., Santos, P.V., Wu, C., Faust, L., de Cock, R., Owens, A.C.S., Lemelin, R.H., Gurung, H., Jusoh, W.F.A., Trujillo, D., Yiu, V., Lopez, P.J., Jaikla, S., Reed, J.M., 2021. Firefly tourism: advancing a global phenomenon toward a brighter future. *Conserv. Sci. Pract.* e391.
- Li, X., Zhou, Y., Zhao, M., Zhao, X., 2020. A harmonized global nighttime light dataset 1992–2018. *Sci. Data* 7, 1–9.
- Lloyd, J.E., 1966. Studies on Flash Communication System in *Photinus* Fireflies. Master's Thesis, University of Michigan, Ann Arbor, Michigan, USA.
- Mänd, R., Tilgar, V., Leivits, A., 2000. Calcium, snails, and birds: a case study. *Web Ecol.* 1, 63–69.
- Mbugua, S.W., Wong, C.H., Ratnayeke, S., 2020. Effects of artificial light on the larvae of the firefly *Lampyris* sp. in an urban city park, peninsular Malaysia. *J. Asia Pac. Entomol.* 23, 82–85.
- McKinley, D.C., Miller-Rushing, A.J., Ballard, H.L., Bonney, R., Brown, H., Cook-Patton, S.C., Evans, D.M., French, R.A., Parrish, J.K., Phillips, T.B., Ryan, S.F., Shanley, L.A., Shirk, J.L., Stepnuck, K.F., Weltzin, J.F., Wiggins, A., Boyle, O.D., Briggs, R.D., Chapin, S.F., Hewitt, D.A., Preuss, P.W., Soukup, M.A., 2017. Citizen science can improve conservation science, natural resource management, and environmental protection. *Biol. Conserv.* 208, 15–28.
- McLean, M., Buck, J., Hanson, F.E., 1972. Culture and larval behavior of photurid fireflies. *Am. Midl. Nat.* 87, 133–145.
- McNeil, D.J., McCormick, E., Heimann, A.C., Kammerer, M., Douglas, M.R., Goslee, S.C., Grozinger, C.M., Hines, H.M., 2020. Bumble bees in landscapes with abundant floral resources have lower pathogen loads. *Sci. Rep.* 10 (1), 22306.
- Mesaglio, T., Soh, A., Kurniawidjaja, S., Sexton, C., 2021. First known photographs of living specimens: The power of iNaturalist for recording rare tropical butterflies. *J. Insect Conserv.* 25, 905–911.
- Müller, J., Hothorn, T., Yuan, Y., Seibold, S., Mitesser, O., Rothacher, J., Freund, J., Wild, C., Wolz, M., Menzel, A., 2023. Weather explains the decline and rise of insect biomass over 34 years. *Nature* 1–6.
- Natural Resources Conservation Service, 2003. National Range and Pasture Handbook. Natural Resources Conservation Service's Grazing Lands Technology Institute, Fort Worth, Texas, USA.
- Neff, F., Korner-Nievergelt, F., Rey, E., Albrecht, M., Bollmann, K., Cahenzli, F., Chittaro, Y., Gossner, M.M., Martinez-Nunes, C., Meier, E.S., Monnerat, C., Moretti, M., Roth, T., Herzog, F., Knop, E., 2022. Different roles of concurring climate and regional land-use changes in past 40 years' insect trends. *Nat. Commun.* 13, 7611.
- Olaya-Arenas, P., Hauri, K., Scharf, M.E., Kaplan, I., 2020. Larval pesticide exposure impacts monarch butterfly performance. *Sci. Rep.* 10, 1–12.
- Oliveira, M.W.D., Trivelin, P.C.O., Boaretto, A.E., Muraoka, T., Mortatti, J., 2002. Leaching of nitrogen, potassium, calcium and magnesium in a sandy soil cultivated with sugarcane. *Pesq. Agrop. Brasileira* 37, 861–868.
- Omernik, J.M., 2004. Perspectives on the nature and definition of ecological regions. *Environ. Manag.* 34, S27–S38.
- Outhwaite, C.L., McCann, P., Newbold, T., 2022. Agriculture and climate change are reshaping insect biodiversity worldwide. *Nature* 605 (7908), 97–102.
- Owens, A.C., Lewis, S.M., 2022. Artificial light impacts the mate success of female fireflies. *R. Soc. Open Sci.* 9, 220468.
- Pearsons, K.A., Lower, S.E., Tooker, J.F., 2021. Toxicity of clothianidin to common eastern north American fireflies. *PeerJ* 9, e12495.
- Picchi, M.S., Avolio, L., Azzani, L., Brombin, O., Camerini, G., 2013. Fireflies and land use in an urban landscape: the case of *Luciola italica* L. (Coleoptera: Lampyridae) in the city of Turin. *J. Insect Conserv.* 17, 797–805.
- Potts, S.G., Biesmeijer, J.C., Kremen, C., Neumann, P., Schweiger, O., Kunin, W.E., 2010. Global pollinator declines: trends, impacts and drivers. *Trends Ecol. Evol.* 25, 345–353.
- PRISM Climate Group, 2014. Oregon State University. <https://prism.oregonstate.edu>.
- Pureswaran, D.S., Roques, A., Battisti, A., 2018. Forest insects and climate change. *Curr. For. Rep.* 4, 35–50.
- R Core Team, 2023. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Raven, P.H., Wagner, D.L., 2021. Agricultural intensification and climate change are rapidly decreasing insect biodiversity. *Proc. Natl. Acad. Sci.* 118, e2002548117.
- Reed, J.M., Nguyen, A., Owens, A.C., Lewis, S.M., 2020. Linking the seven forms of rarity to extinction threats and risk factors: an assessment of north American fireflies. *Biodivers. Conserv.* 29, 57–75.
- Ridenhour, K., 2022. Atlanta Firefly Project: Using Community Science to Explore the Effects of Land Management on the Local Abundance of the Big Dipper Firefly (*Photinus pyralis*) in City Parks and Residential Yards. Doctoral dissertation, University of Georgia, Athens, Georgia, USA.

- Rosenberg, K.V., Will, T., Buehler, D.A., Swarthout, S.B., Thogmartin, W.E., Bennett, R.E., Chandler, R., 2016. Dynamic distributions and population declines of Golden-winged warblers. *Stud. Avian Biol.* 49, 3–28.
- Sambaraju, K.R., Carroll, A.L., Zhu, J., Stahl, K., Moore, R.D., Aukema, B.H., 2012. Climate change could alter the distribution of mountain pine beetle outbreaks in western Canada. *Ecography* 35, 211–223.
- Sands, D.P., 2018. Important issues facing insect conservation in Australia: now and into the future. *Austral Entomol.* 57, 150–172.
- Schulte, L.A., Niemi, J., Helmers, M.J., Liebman, M., Arbuckle, J.G., James, D.E., Witte, C., 2017. Prairie strips improve biodiversity and the delivery of multiple ecosystem services from corn-soybean croplands. *Proc. Natl. Acad. Sci.* 114 (42), 11247–11252.
- Spinu, V., Grolemond, G., Wickham, H., Lyttle, I., Costigan, I., Law, J., Mitarotonda, D., Larmarange, J., Boiser, J., Lee, C.H., 2021. Lubridate: making dealing with dates a little easier. R Package. Version 1 (7), 10.
- Stepanian, P.M., Entekhin, S.A., Wainwright, C.E., Mirkovic, D., Tank, J.L., Kelly, J.F., 2020. Declines in an abundant aquatic insect, the burrowing mayfly, across major north American waterways. *Proc. Natl. Acad. Sci.* 117, 2987–2992.
- Sullivan, B.L., Aycrigg, J.L., Barry, J.H., Bonney, R.E., Bruns, N., Cooper, C.B., Kelling, S., 2014. The eBird enterprise: an integrated approach to development and application of citizen science. *Biol. Conserv.* 169, 31–40.
- Teucher, A., Rudis, B., 2019. Lutz: look up time zones of point coordinates. R package version 0 (3), 1.
- Thomas, J.A., 2016. Butterfly communities under threat. *Science* 353, 216–218.
- Tulloch, A.I., Possingham, H.P., Joseph, L.N., Szabo, J., Martin, T.G., 2013. Realising the full potential of citizen science monitoring programs. *Biol. Conserv.* 165, 128–138.
- US Geologic Survey, 2019. 3DEP Digital Elevation Model. <https://www.usgs.gov/3d-elevation-program>.
- USDA National Agricultural Statistics Service, 2016. Cropland data layer. Published crop-specific data layer [online]. Available at: <https://croplandcros.scinet.usda.gov/>.
- USDA Natural Resources Conservation Service, Soil Survey Staff, 2020. Gridded Soil Survey Geographic (gSSURGO) Database for the Conterminous United States. <https://gdg.sc.egov.usda.gov/>.
- Van Klink, R., Bowler, D.E., Gongalsky, K.B., Swengel, A.B., Gentile, A., Chase, J.M., 2020. Meta-analysis reveals declines in terrestrial but increases in freshwater insect abundances. *Science* 368, 417–420.
- Van Swaay, C.A.M., Dennis, E.B., Schmucki, R., Sevilleja, C.G., Åström, S., Balalaikins, M., Barea-Azcón, J.M., Bonelli, S., Botham, Roy, D.B., 2022. European Grassland Butterfly Indicator 1990-2020 Technical report. Butterfly Conservation Europe & SPRING/eBMS. www.butterfly-monitoring.net. & Vlinderstichting report VS2022.039.
- Viviani, V.R., Rosa, S.P., Martins, M.A., 2012. *Aspisoma lineatum* (Gyllenhal) (Coleoptera: Lampyridae) firefly: description of the immatures, biological, and ecological aspects. *Neotrop. Entomol.* 41, 89–94.
- Wagner, D.L., 2020. Insect declines in the Anthropocene. *Annu. Rev. Entomol.* 65, 457–480.
- Wagner, D.L., Van Driesche, R.G., 2010. Threats posed to rare or endangered insects by invasions of nonnative species. *Annu. Rev. Entomol.* 55, 547–568.
- Wagner, D.L., Fox, R., Salcido, D.M., Dyer, L.A., 2021a. A window to the world of global insect declines: moth biodiversity trends are complex and heterogeneous. *Proc. Natl. Acad. Sci.* 118 (2), e2002549117.
- Wagner, D.L., Grames, E.M., Forister, M.L., Berenbaum, M.R., Stopak, D., 2021b. Insect decline in the Anthropocene: death by a thousand cuts. *Proc. Natl. Acad. Sci.* 118, e2023989118.
- Wang, A., Goslee, S.C., Miller, D.A., Sanderson, M.A., Gonet, J.M., 2017. Topographic variables improve climatic models of forage species abundance in the northeastern United States. *Appl. Veg. Sci.* 20, 84–93.
- Wepprich, T., Adrion, J.R., Ries, L., Wiedmann, J., Haddad, N.M., 2019. Butterfly abundance declines over 20 years of systematic monitoring in Ohio, USA. *PLoS One* 14 (7), e0216270.
- Wilcove, D.S., Rothstein, D., Dubow, J., Phillips, A., Losos, E., 1998. Quantifying threats to imperiled species in the United States. *BioScience* 48, 607–615.
- Wright, M.N., Ziegler, A., 2017. Ranger: a fast implementation of random forests for high dimensional data in C++ and R. *J. Stat. Softw.* 77, 1–17.
- Wright, M.N., Wager, S., Probst, P., 2020. Ranger: a fast implementation of random forests. R package version 0 (12), 1.
- Xian, G., Homer, C., Dewitz, J., Fry, J., Hossain, N., Wickham, J., 2011. The change of impervious surface area between 2001 and 2006 in the conterminous United States. *Photogramm. Eng. Remote. Sens.* 77, 758–762.
- Yang, L., Huang, C., Homer, C., Wylie, B., Coan, M., 2003. An approach for mapping large-area impervious surfaces: synergistic use of Landsat 7 ETM+ and high spatial resolution imagery. *Can. J. Remote. Sens.* 29, 230–240.
- Ziesche, T.M., Ordon, F., Schliephake, E., Will, T., 2023. Long-term data in agricultural landscapes indicate that insect decline promotes pests well adapted to environmental changes. *J. Pest. Sci.* 1–17.